PROJECT AND TEAM INFORMATION

Project Title
Autonomous Drone Fleet Routing Management System

Student / Team Information	
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PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

Last-mile delivery represents the most expensive, carbon-intensive, and congestion-prone component of modern logistics, often exceeding 50% of total shipping costs[cite: 1]. Ground courier fleets encounter persistent traffic gridlock, non-linear transit delays, and geographical barriers[cite: 1, 3]. Autonomous Unmanned Aerial Vehicles (UAVs) provide a transformative paradigm by utilizing low-altitude airspace for point-to-point delivery at sustained cruise speeds[cite: 1, 3]. However, operating a multi-drone fleet introduces strict constraints: bounded lithium-polymer battery capacities, non-linear aerodynamic energy expenditure with payload weight and wind vectors, and tight customer deadlines[cite: 1, 2]. Solving this problem is vital for automated logistics in e-commerce, urgent healthcare, and emergency disaster relief[cite: 1, 3].

State of the Art / Current solution (1 pt)

Current logistics relies predominantly on manual ground delivery or single-drone systems operating from a central depot (SD)[cite: 1]. While recent research has formulated collaborative vehicle-drone models (FSTSP and VRPD), most practical frameworks simplify flight physics by assuming static energy consumption rates and zero ambient wind disturbance[cite: 1, 2]. Furthermore, existing dispatch systems lack modular data structures for real-time order prioritization, dynamic re-routing, and synchronized battery swapping with minimal runtime latency[cite: 1, 2].

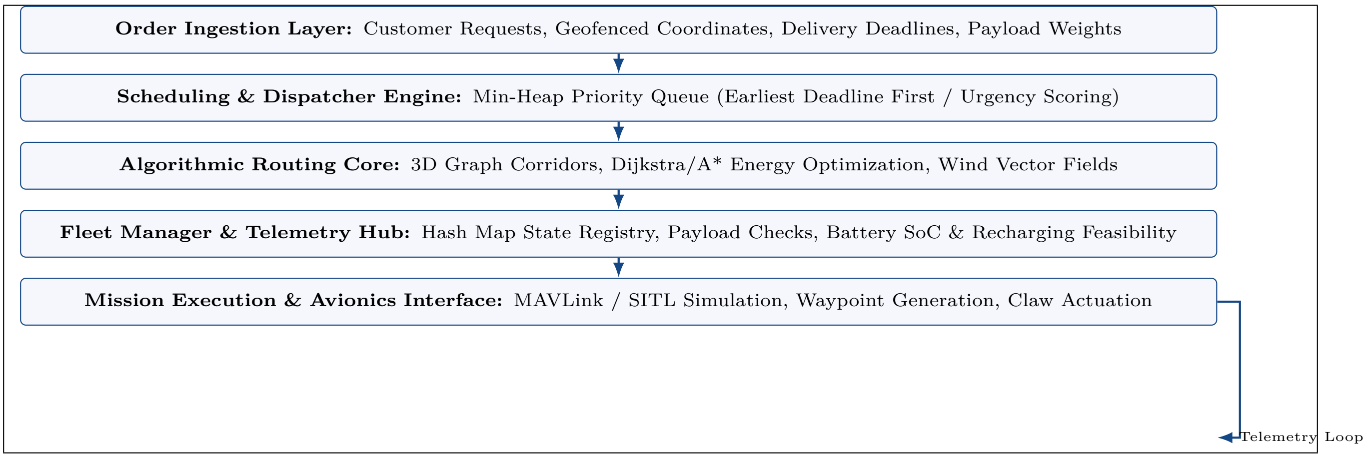
Project Goals and Milestones (2 pts)

The goal is to develop a high-performance simulation and routing management engine in modern C++ using optimized Data Structures and Algorithms (DSA)[cite: 1, 2]. **Key Milestones:**
• **Milestone 1 (Formulation):** Formalize the mathematical MILP, time windows, and aerodynamic power models with wind vector dynamics[cite: 1, 2].
• **Milestone 2 (Core DSA Engine):** Implement Min-Heap Priority Queues for dynamic scheduling, Graph Adjacency Lists, and Hash Tables for $O(1)$ fleet telemetry.
• **Milestone 3 (Algorithmic Routing):** Implement Dijkstra and A* search over 3D airway corridors with dynamic edge weight re-calculation under wind fields[cite: 1].
• **Milestone 4 (Integration):** Assemble the Object-Oriented software engine, validate mission feasibility, and benchmark makespan reduction[cite: 1, 2].

Project Approach (3 pts)

The system is architected as an Object-Oriented C++17 application structured around modular layers. Customer delivery requests are ingested and sorted in an Earliest-Deadline-First (EDF) Min-Heap Priority Queue. The fleet management subsystem maintains drone telemetry in an unordered hash map for average $O(1)$ state queries. 3D airspace corridors are represented as a directed weighted graph using an adjacency list. The pathfinding module calculates energy-optimal flight paths using Dijkstra's algorithm, where edge weights dynamically adapt to aerodynamic drag and ambient wind vectors ($\vec{v}_g = \vec{v}_a + \vec{v}_w$)[cite: 1]. The system is built using GCC/Clang on Linux, formatted with CMake, and documented through Overleaf $\LaTeX$.

System Architecture (High Level Diagram) (2 pts)



Project Outcome / Deliverables (1 pts)

The deliverable is a production-grade, compilable C++17 autonomous drone fleet management and routing software package. Deliverables include: (1) Core DSA modules implementing Min-Heap priority queues, graph adjacency lists, and hash table fleet tracking; (2) Aerodynamic energy and wind-field computation routines; (3) Dynamic order assignment and Dijkstra 3D trajectory calculation modules; and (4) Comprehensive experimental benchmarking datasets demonstrating fleet makespan and energy reduction[cite: 1, 2].

Assumptions

Drones cruise at constant airspeed within designated altitude corridors; weather conditions and wind velocity vectors remain quasi-static during individual flight legs; each customer order is serviced by a single dedicated drone flight leg; and battery charging/swapping resets battery State of Charge (SOC) to 100% at designated depot stations[cite: 1].

References

[1] G. Attenni, V. Arrigoni, N. Bartolini, and G. Maselli, "Drone-based Delivery Systems: A survey on Route Planning," *IEEE Access*, vol. 11, pp. 123476–123504, 2023.
[2] J. Kong, L. Wei, and X. Jiang, "A Literature Review of Vehicle and Drone Delivery Routing Problems in Different Synchronization Level Scenarios," *Drones*, vol. 10, no. 3, p. 206, MDPI, 2026.
[3] S. Pathan, H. Pandey, A. Savare, and Prof. N. Dumne, "Autonomous Drone Delivery System," *IJARCCE*, vol. 14, no. 11, pp. 236–239, 2025.
[4] C. C. Murray and A. G. Chu, "The flying sidekick traveling salesman problem," *Transp. Res. Part C*, vol. 54, pp. 86–109, 2015.